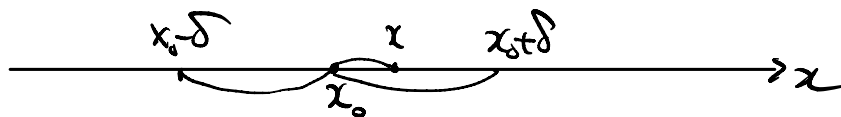


二. 多元函数之极限. (∞ = 元函数为主).

一元函数 $f(x)$ 在 x_0 之极限: ε - δ 语言.

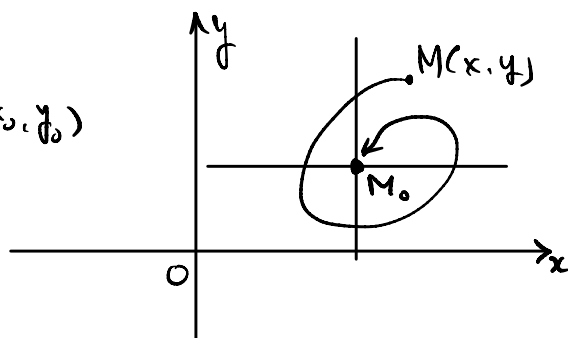
$$\forall \varepsilon > 0, \exists \delta > 0, \Rightarrow \forall x: 0 < |x - x_0| < \delta \text{ 有 } |f(x) - A| < \varepsilon$$



二元函数 $f(x, y)$: $M(x, y), M_0(x_0, y_0)$

$$r(M, M_0) \rightarrow 0: f(M) \rightarrow A$$

按任意方向



定义 2.2. $M \in \mathbb{C} \mathbb{R}^2$. $M_0(x_0, y_0)$ 是 E 之聚点. $A \in \mathbb{R}$.

$$\text{如果 } \forall \varepsilon > 0, \exists \delta > 0, \Rightarrow \forall M \in E: 0 < r(M, M_0) < \delta$$

$$|f(M) - A| < \varepsilon \quad (\forall M \in O(M_0, \delta) \cap E)$$

那么称当 $M \rightarrow M_0$ 时, $f(M)$ 收敛于 A , 称 A 是 $M \rightarrow M_0$ 时 $f(M)$ 之极限. 记为

$$\lim_{M \rightarrow M_0} f(M) = A, \quad \lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A \quad \text{或} \quad \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) = A$$

$$\text{或 } f(x, y) \rightarrow A \quad (x, y) \rightarrow (x_0, y_0).$$

注: ① M_0 是 E 之聚点: $\forall \delta > 0, O(M_0, \delta) \cap E \neq \emptyset$.
 $f(M)$ 在 M_0 不必有定义.

$$\textcircled{2} \quad \underline{M \rightarrow M_0}: f(M) \rightarrow A \quad M \rightarrow M_0 \text{ 之方式}$$

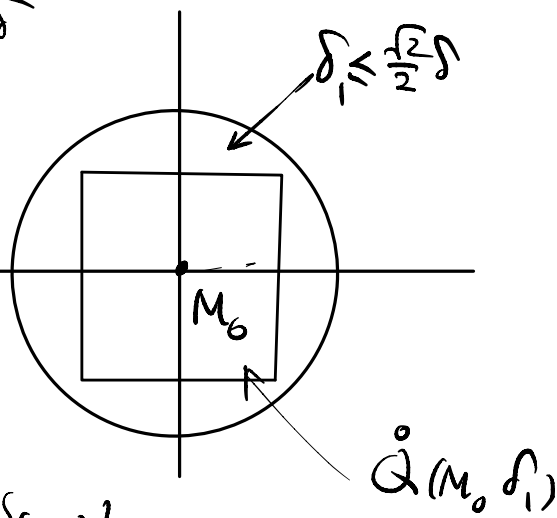
是任意之, 只要 $r(M, M_0) \rightarrow 0$

$$③ \quad M \in E \quad \& \quad 0 < r(M, M_0) < \delta$$

即有:

$$\overset{\circ}{Q}(M_0, \delta)$$

$$\text{即有} \begin{cases} |x-x_0| < \delta_1 \\ |y-y_0| < \delta_1 \end{cases} \quad \& \quad (x, y) \neq (x_0, y_0)$$

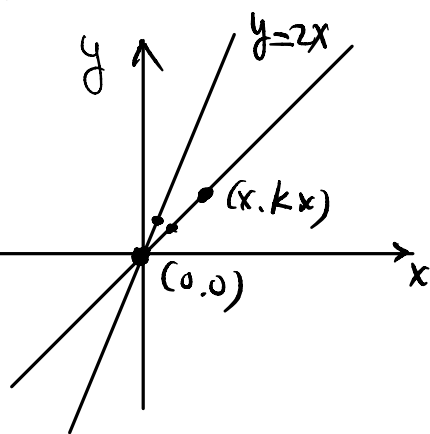


$$\text{例. } f(x, y) = \frac{xy}{x^2+y^2}, \quad E = \mathbb{R}^2 \setminus \{(0,0)\}$$

$$M_0(0,0): \quad M(x,y) \rightarrow M_0(0,0):$$

$$\frac{1}{2} \quad y=kx \xrightarrow{x \rightarrow 0} 0:$$

$$k \in \mathbb{R} \quad f(x, kx) = \frac{xy}{x^2+y^2} = \frac{x \cdot kx}{x^2+k^2x^2} = \frac{k}{1+k^2}$$



$$k=1: \quad f(x, x) = \frac{1}{2}; \quad = A_1$$

$$k=2: \quad f(x, 2x) = \frac{2}{1+2^2} = \frac{2}{5} = A_2 \quad A_1 \neq A_2$$

$(x, y) \rightarrow (0,0):$ $f(x, y)$ 极限不存在!
 $f(x, y)$ 不收敛.

$$\text{例} \quad f(x, y) = \frac{x^2y}{x^4+y^2}, \quad \mathbb{R}^2 \setminus \{(0,0)\}$$

$$(x, y) \rightarrow (0,0): \quad f(x, y) \rightarrow ?$$

$$y=kx^2 \rightarrow 0: \quad f(x, kx^2) = \frac{x^2 \cdot kx^2}{x^4 + k^2x^4} = \frac{k}{1+k^2} \quad \text{常数}$$

$$y=x^2 \rightarrow 0: \quad f(x, x^2) = \frac{1}{2}; \quad = A_1$$

$$A_1 \neq A_2:$$

$$y=2x^2 \rightarrow 0: \quad f(x, 2x^2) = \frac{2}{5} = A_2$$

$(x, y) \rightarrow (0,0) \quad f(x, y)$ 不收敛; $\lim_{(x,y) \rightarrow (0,0)} f(x, y) \neq$

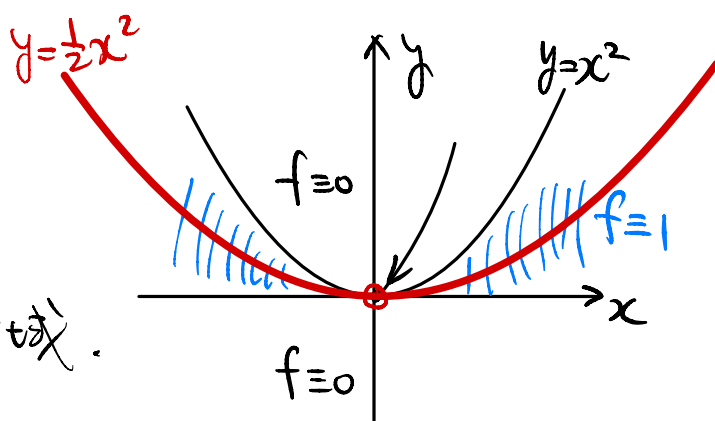
例 $f(x,y) = \begin{cases} 1, & 0 < y < x^2 \\ 0, & \text{其他} \end{cases}$

$E = \{(x,y) \mid 0 < y < x^2\}$: 非空
不是区域.

$f(x,y) = 1_E(x,y) = \chi_E(x,y)$

$= \begin{cases} 1, & (x,y) \in E \\ 0, & (x,y) \notin E \end{cases}$

E 的特征函数 (Characteristic function.)



χ : chi

$(0,0)$: $(x,y) \rightarrow (0,0)$: $(x,y) \in$ 下平面: $(x,y) \notin E$;
 $y < 0$,
 $f(x,y) \equiv 0$.

$y < 0$, $\begin{cases} (x,y) \notin E; f(x,y) \equiv 0; \end{cases}$ $(x,y) \rightarrow (0,0)$ 时: $f(x,y) \rightarrow 0$;

$0 < y = \frac{1}{2}x^2 < x^2$: $(x,y) \rightarrow (0,0)$ $f(x,y) \equiv 1 \rightarrow 1$
 $x \neq 0$, $\in E$;

$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) \nexists$. $\nexists$, $(x,y) \rightarrow (0,0)$ 时 $f(x,y)$ 不收敛.

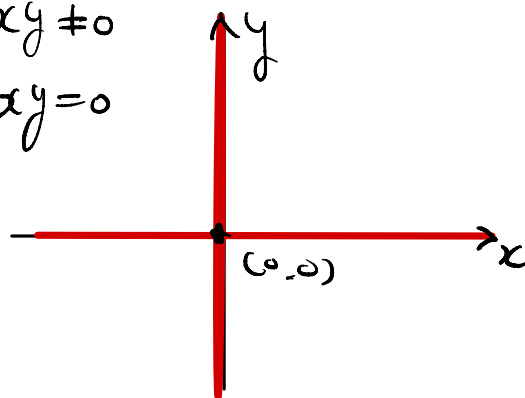
例 $f(x,y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x}, & xy \neq 0 \\ 0, & xy = 0 \end{cases}$

$(x,y) \rightarrow (0,0)$:

$|f(x,y)| \leq |x| + |y| \rightarrow 0$
 $(x,y) \rightarrow (0,0)$

$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

$y \neq 0$: $\lim_{x \rightarrow 0} f(x,y) \nexists$; $x \neq 0$: $\lim_{y \rightarrow 0} f(x,y) \nexists$.



一元函数的 Heine 定理:

$$\lim_{x \rightarrow x_0} f(x) = A \iff \forall \{x_n\}: x_n \rightarrow x_0, (x_n \neq x_0), f(x_n) \rightarrow A.$$

二元函数极限的 Heine 定理:

$f(x, y)$ 定义于 $E \subset \mathbb{R}^2$, $M_0(x_0, y_0)$ 是 E 的聚点. 则

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A \iff \forall \text{点列 } M_k(x_k, y_k) \in E \setminus \{M_0\}, M_k \rightarrow M_0, \text{ 有 } f(M_k) \rightarrow A. (k \rightarrow \infty)$$

证明: 设 $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A$.

$$\forall \varepsilon > 0, \exists \delta > 0, \Rightarrow \forall M(x, y) \in E: 0 < r(M, M_0) < \delta: \text{ 有 } |f(M) - A| < \varepsilon.$$

$$\forall \text{点列 } M_k \in E \setminus \{M_0\}: 0 < r(M_k, M_0) \xrightarrow{k \rightarrow \infty} 0$$

$$\text{对于上述 } \delta: \exists N > 0, \Rightarrow \forall k > N: 0 < \underline{r(M_k, M_0)} < \delta$$

$$\therefore \forall k > N: M_k: |f(M_k) - A| < \varepsilon$$

$$\therefore f(M_k) \rightarrow A \quad (k \rightarrow \infty)$$

反之: 设 $\forall \text{点列 } M_k \in E \setminus \{M_0\}, M_k \rightarrow M_0$, 都有 $f(M_k) \rightarrow A$.

$$\text{求证: } \lim_{M \rightarrow M_0} f(M) = A.$$

用反证法: “ $\lim_{M \rightarrow M_0} f(M) = A$ ” 不成立:

$$\exists \varepsilon_0 > 0 \Rightarrow \forall \delta > 0: \exists M_r \in E \setminus \{M_0\}, \text{ 尽管 } r(M_r, M_0) < \delta:$$

$$|f(M_r) - A| \geq \varepsilon_0.$$

$$\delta = \frac{1}{k}: \exists M_k \in E \setminus \{M_0\}: \underline{r(M_k, M_0)} < \frac{1}{k}: \text{ 但 } |f(M_k) - A| \geq \varepsilon_0.$$

$$0 < r(M_k, M_0) < \frac{1}{k} \rightarrow 0: \quad \therefore M_k \rightarrow M_0 \quad (M_k \neq M_0):$$

由大 ϵ 提: $f(M_k) \rightarrow A$.

这与 " $|f(M_k) - A| \geq \epsilon_0 > 0$ " 矛盾!

$$\therefore \lim_{M \rightarrow M_0} f(M) = A. \quad \#$$

$$\text{例: } f(x, y) = \chi_E(x, y) \quad E = \{(x, y) \mid 0 < y < x^2\}.$$

$$\left. \begin{array}{l} y < 0, (x, y) \rightarrow (0, 0), f(x, y) \equiv 0 \rightarrow 0 \\ 0 < y = \frac{1}{2}x^2, (x, y) \rightarrow (0, 0), f(x, y) \equiv 1 \rightarrow 1 \end{array} \right\} \text{不收敛,}$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \nexists$$

$$\text{例 } f(x, y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x}, & xy \neq 0 \\ 0, & xy = 0 \end{cases}$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0; \quad \text{但: } \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y) \neq \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y) \neq$$

三. 重极限与累次极限 (= 次极限).

$$\begin{array}{ccc} M(x, y) & M_0(x_0, y_0) & \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} \\ \text{重极限: } \lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A, & \text{累次极限:} & \end{array}$$

$$\forall \epsilon > 0, \exists \delta > 0, \Rightarrow \forall M(x, y) \in E \setminus \{M_0\}, 0 < r(M, M_0) < \delta \\ |f(M) - A| < \epsilon.$$

定2.4. 如果 $\forall y \neq y_0$ 有: $\lim_{x \rightarrow x_0} f(x, y) = \varphi(y)$ 且

$\lim_{y \rightarrow y_0} \varphi(y) = A$: 那么 A 是 $f(x, y)$ 在 $x \rightarrow x_0$ 后 $y \rightarrow y_0$

在累次极限. 即 $\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) = A$.
(= 次)

同理: 固定 $x \neq x_0$. $\psi(x) = \lim_{y \rightarrow y_0} f(x, y)$, $\lim_{x \rightarrow x_0} \psi(x) = B$.

称 B 是 $f(x, y)$ 在 $y \rightarrow y_0$ 后 $x \rightarrow x_0$ 的累次极限. 即
 $\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = B$. (= 次)

例. $f(x, y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x}, & xy \neq 0 \\ 0, & xy = 0 \end{cases}$

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$ 重极限存在;

但: $\lim_{x \rightarrow 0} f(x, y) \nexists$ $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right) \nexists$
累次极限不存在. $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y) \nexists$

例 $f(x, y) = \frac{xy}{x^2 + y^2}$:

在 $(0, 0)$: 重极限不存在. $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{x^2 + y^2} \nexists$,

$y \neq 0$: $\lim_{x \rightarrow 0} f(x, y) = 0$; $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} f(x, y) \right) = 0$

$x \neq 0$: $\lim_{y \rightarrow 0} f(x, y) = 0$. $\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} f(x, y) \right) = 0$.

累次极限存在且相等. 但重极限不存在.

例 $f(x, y) = \frac{x - y + x^2 + y^2}{x + y}$.

$\because x = y \rightarrow 0$: $f(x, x) = \frac{2x^2}{2x} = x \xrightarrow{x \rightarrow 0} 0$

$x = 2y \rightarrow 0$: $f(x, 2y) = \frac{2y - y + y^2}{3y} = \frac{1}{3} + \frac{1}{3}y \rightarrow \frac{1}{3}$ 不相等.

$\therefore (x, y) \rightarrow (0, 0)$ 重极限 $\nexists$

$$y \neq 0: \lim_{x \rightarrow 0} f(x, y) = \frac{-y+y^2}{y} = -1+y.$$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y) = -1;$$

$$x \neq 0: \lim_{y \rightarrow 0} f(x, y) = \frac{x+x^2}{x} = 1+x$$

累次极限 $\exists$

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y) = 1.$$

但不相等.

例 $\lim_{(x,y) \rightarrow (0,0)} \frac{y}{x} \nexists$.

$$x \neq 0 \quad \lim_{y \rightarrow 0} \frac{y}{x} = 0; \quad \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{y}{x} \right) = 0$$

$$y \neq 0. \quad \lim_{x \rightarrow 0} \frac{y}{x} \nexists \quad \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{y}{x} \nexists$$

定理 2.5. 若重极限 $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y) = A$ 存在, 且 $\exists \delta_0 > 0$

$$\Rightarrow \forall y \neq y_0: |y - y_0| < \delta_0: \lim_{x \rightarrow x_0} f(x, y) = \varphi(y) \exists,$$

$$\text{则累次极限 } \lim_{y \rightarrow y_0} \varphi(y) = \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) = A.$$

证明: $\because$ 重极限 $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y) = A$ 存在,

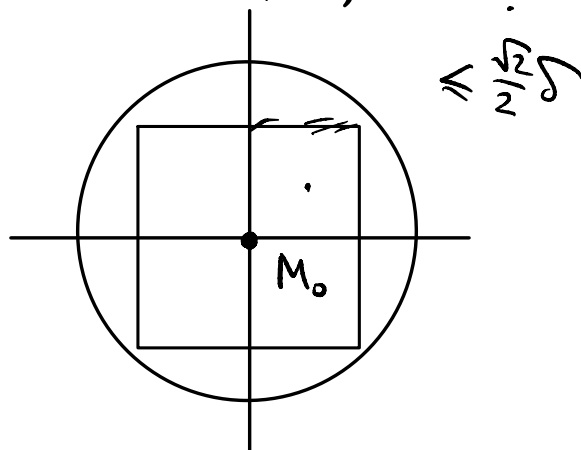
$$\therefore \forall \varepsilon > 0, \exists \delta_1 > 0, \exists \forall M(x, y): 0 < \rho(M, M_0) < \delta_1 \text{ 时},$$

$$|f(x, y) - A| < \varepsilon.$$

$$\forall x, \quad 0 < |x - x_0| < \frac{\sqrt{2}}{2} \delta_1 \triangleq \delta,$$

$$\forall y: \quad 0 < |y - y_0| < \frac{\sqrt{2}}{2} \delta_1 = \delta$$

$$\Rightarrow |f(x, y) - A| < \varepsilon.$$



对 $\delta \in \delta = \frac{\sqrt{2}}{2} \delta_1$: $\forall y, 0 < |y - y_0| < \delta$, 由

$$\begin{array}{c} |f(x, y) - A| < \varepsilon \\ \downarrow \\ | \varphi(y) - A | \leq \varepsilon \end{array}$$

令 $x \rightarrow x_0$ 得

$$\therefore \lim_{y \rightarrow y_0} \varphi(y) = A.$$

即 $\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) = A$. #

推论 2.6. 若重极限与累次极限都存在, 必相等.

推论 2.7. 若累次极限都存在, 但不相等, 则重极限不存在.